

EXPERIMENT 40

Launch Hadoop 2.x and Test the MapReduce Platform

AIM

To launch the configured Hadoop 2.x single-node cluster and test the MapReduce platform by running a sample word-count job.

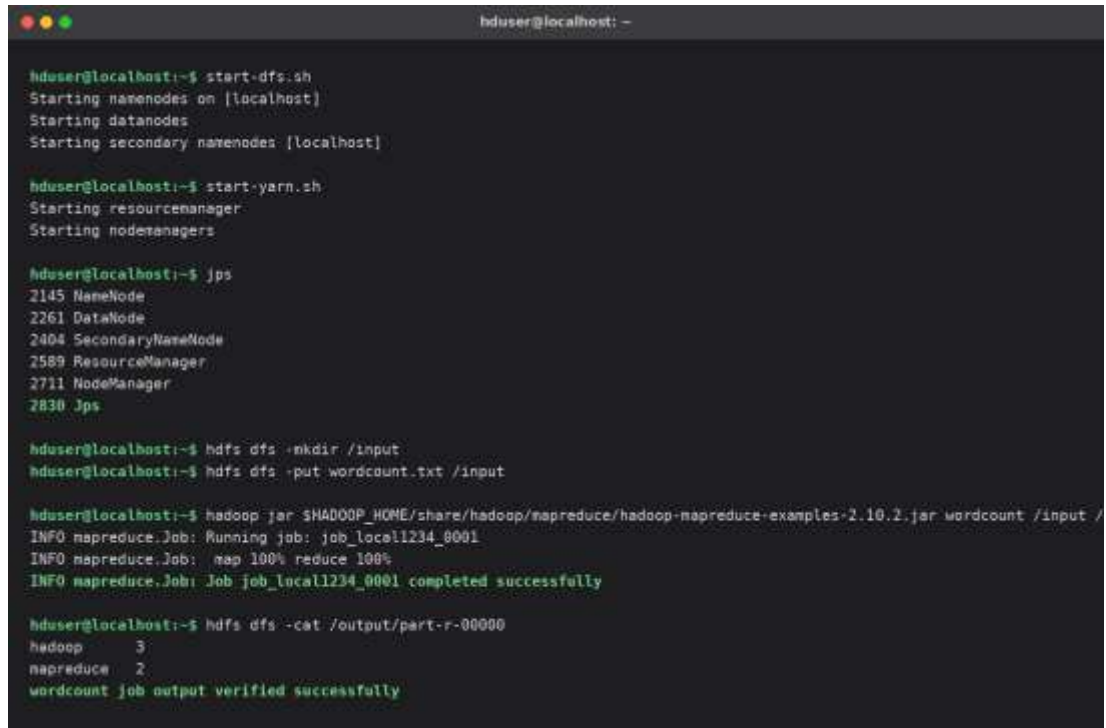
REQUIREMENTS

- Hadoop 2.x installed and configured (see Experiment 39)
- HDFS NameNode already formatted
- hduser account with passwordless SSH to localhost
- Sample input text file for the MapReduce word-count job

PROCEDURE

1. Start the HDFS daemons (NameNode, DataNode, Secondary NameNode) using start-dfs.sh.
2. Start the YARN daemons (ResourceManager, NodeManager) using start-yarn.sh.
3. Run the jps command to verify that all required Hadoop daemons are running.
4. Create an input directory in HDFS using the `hdfs dfs -mkdir` command.
5. Upload a sample text file into the HDFS input directory using `hdfs dfs -put`.
6. Run the bundled word-count MapReduce example jar against the input directory.
7. Monitor the job progress until the map and reduce phases reach 100 percent completion.
8. View the job output using the `hdfs dfs -cat` command on the output directory.
9. Stop the daemons using stop-yarn.sh and stop-dfs.sh once testing is complete.

IMPLEMENTATION

A terminal window titled 'hduser@localhost: -' showing the process of starting a Hadoop cluster and running a word-count job. The user enters several commands: 'start-dfs.sh' to start the HDFS daemons, 'start-yarn.sh' to start the YARN daemons, 'jps' to list the active processes, 'hdfs dfs -mkdir /input' to create an input directory, 'hdfs dfs -put wordcount.txt /input' to upload the test file, 'hadoop jar \$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-2.10.2.jar wordcount /input /' to run the word-count job, and 'hdfs dfs -cat /output/part-r-00000' to view the output. The terminal output shows the successful startup of NameNode, DataNode, SecondaryNameNode, ResourceManager, and NodeManager, followed by the successful completion of the word-count job with verified output.

```
hduser@localhost:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [localhost]

hduser@localhost:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers

hduser@localhost:~$ jps
2145 NameNode
2261 DataNode
2404 SecondaryNameNode
2589 ResourceManager
2711 NodeManager
2830 Jps

hduser@localhost:~$ hdfs dfs -mkdir /input
hduser@localhost:~$ hdfs dfs -put wordcount.txt /input

hduser@localhost:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-2.10.2.jar wordcount /input /
INFO mapreduce.Job: Running job: job_local1234_0001
INFO mapreduce.Job: map 100% reduce 100%
INFO mapreduce.Job: Job job_local1234_0001 completed successfully

hduser@localhost:~$ hdfs dfs -cat /output/part-r-00000
hadoop      3
mapreduce   2
wordcount job output verified successfully
```

Figure 1: Illustrative terminal view — Hadoop cluster startup and MapReduce word-count test.

RESULT

The Hadoop 2.x cluster was launched successfully with all daemons active, and the sample MapReduce word-count job completed successfully with verified output.